Information-Aware Multiobjective Evolutionary Optimization with Multi-Fidelity Evaluation for UAV-Assisted Federated Learning

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ABSTRACT

This working document develops a UAV-assisted federated learning problem in which client selection must account for complementary data, heterogeneous communication resources, and incomplete information. Ground devices train locally, UAVs relay and aggregate updates, and a coordinator selects participating devices and allocates bandwidth and computation resources. Three objectives are considered: minimize population validation loss, total operational energy, and elapsed training time. The document separates a reference simulator benchmark from an implementable controller that only observes historical reports and explicitly acquired information. Three algorithmic directions are specified: (A) set-aware multiobjective evolutionary scheduling, (B) multi-fidelity evolutionary optimization with cost-aware evaluation and a verified Pareto archive, and (C) optional adaptive probing with two-stage online recourse. Mathematical assumptions, numerical examples, encodings, pseudocode, experimental controls, student work packages, and verification exercises are provided. Evaluation fidelity is an optimization mechanism; physical probing is a network action with its own cost. Proposed contributions and hypotheses require empirical validation and further novelty assessment before journal submission.

1. System model and motivating learning scenario

1.1. Research question and scope

Consider geographically distributed sensors or cameras supported by UAV-mounted edge servers. Most clients observe common conditions, while a few observe rare but important events. Good wireless channels do not necessarily identify useful training data. Conversely, a large deviation from the average gradient can indicate useful minority information or corrupted observations. The main research question is:

> How can a scheduler select complementary clients and allocate wireless resources when evaluating candidate schedules is expensive and current client information is only partially available?

The physical problem, learning objective, and optimization algorithm must remain distinct. A cheap proxy can guide search but must not silently replace the population-level learning objective.

The initial model uses prescribed UAV movement, orthogonal spectrum pools, synchronous FL, fixed local SGD steps and minibatch size, fixed uplink transmit power, and stationary datasets. Independent application offloading, optimized flight trajectories, personalized models, and concept drift are extensions. Direction A establishes a reference-evaluated scheduler; Direction B addresses search cost; Direction C introduces adaptive information acquisition. Students should implement these layers sequentially.

1.2. Network, clocks, and observable information

Let $\mathcal{M} = \{1, \dots, M\}$ be the clients and $\mathcal{U} = \{1, \dots, U\}$ the UAVs. A coordinator aggregates UAV outputs. Round $k \in \{0, \dots, K-1\}$ starts with global model $w^k \in \mathbb{R}^d$ at physical time τ_k. Positions are $q_u(\tau_k) \in \mathbb{R}^3$ and $p_m(\tau_k) \in \mathbb{R}^3$. The clock follows

$$\tau_{k+1} = \tau_k + T_k, \tag{1}$$

where T_k includes decision-making and service time. Prescribed motion is indexed by physical time, not by the algorithm's FL round number.

A first simulator freezes each channel within a round. This is a block-channel approximation; its suitability must be checked against round duration and mobility. An extension updates channels within a round. The controller observes

ORCID(s):

Table 1
Core notation and units.

Symbol	Meaning	Unit
w^k, Δ_m^k	Global parameters and client model increment	model-dependent
p_m	Fixed client weight in the target population	dimensionless
$s_{m,u}^k, z_{m,u}^k$	Training and probing associations	binary
$b_{m,u}^{r,k}, f_m^{r,k}$	Bandwidth and device CPU frequency	Hz, cycles/s
$C_m^{r,k}, L_m^r$	CPU work and upload payload, $r \in \{P, T\}$	cycles, bits
$P_m, N_0, h_{m,u}^k$	Uplink power, noise PSD, channel gain	W, W/Hz, dimensionless
E_k, T_k	Executed energy and duration per round	J, s
$\mathcal{H}_k, a_m^k$	Available history and information age	history, s
$\widehat{\mathbf{j}}_\ell(x), c_\ell(x)$	Fidelity estimate and evaluation cost	vector, GPU-s or s

measured links, hardware limits, batteries, and authorized learning reports. It does not automatically observe all current gradients or labels. Hidden simulator fields are restricted to explicitly identified oracle evaluations.

1.3. A small motivating scenario

Start with two UAVs and six clients. Four clients observe common classes, one observes a rare class, and one has noisy labels. The rare-class client has a weak link. Choosing only nearby clients can neglect the rare class; choosing all unusual gradients can favor noise. A useful method must assess client sets and their resource cost together. The six-client setting supports restricted enumeration and should be understood before increasing scale.

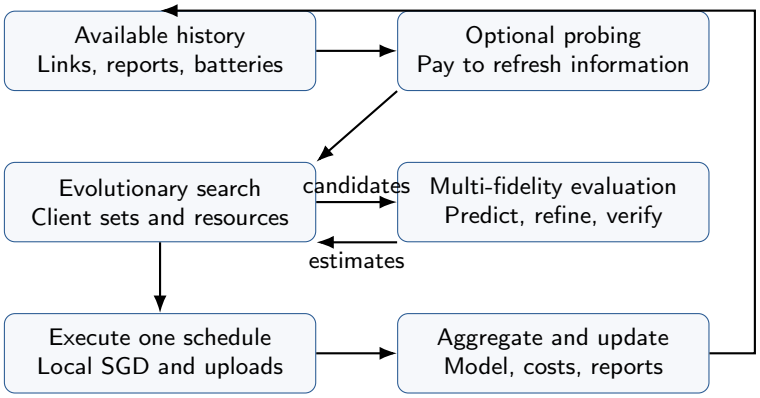


Figure 1: Conceptual controller. Physical probing changes the available information and consumes operational resources. Multi-fidelity evaluation guides search; simulated candidate trials are not free physical FL rounds.

2. Federated learning, complementary data, and uncertainty

2.1. Population objective and local training

Client m holds training set $\mathcal{D}_m$ with n_m samples. Define

$$F_m(w) = \frac{1}{n_m} \sum_{\xi \in \mathcal{D}_m} \ell(w; \xi), \qquad F_{\text{pop}}(w) = \sum_{m=1}^{M} p_m F_m(w), \quad p_m > 0, \quad \sum_m p_m = 1. \tag{2}$$

Sample weighting uses $p_m = n_m / \sum_j n_j$; device balancing uses $p_m = 1/M$. Region-based weights are possible if defined by the application in advance. The target weights must not change just because a scheduler excludes difficult clients.

Let $\mathcal{V}_m$ be held-out validation data. The optimization target is

$$L_{\mathrm{val}}(w) = \sum_m p_m \frac{1}{|\mathcal{V}_m|} \sum_{\xi \in \mathcal{V}_m} \ell(w; \xi). \tag{3}$$

Training, validation, and final test data are disjoint. Equation (3) does not assume that a real coordinator holds all validation records. A simulator can evaluate it centrally; deployment requires an authorized evaluation service or distributed validation with explicitly charged overhead.

Selected client m performs

$$w^k_{m,0} = w^k, \tag{4}$$

$$w^k_{m,i+1} = w^k_{m,i} - \eta_k \widehat{\nabla} F_m(w^k_{m,i}), \quad i = 0, \dots, I-1, \tag{5}$$

$$\Delta^k_m = w^k_{m,I} - w^k. \tag{6}$$

Fix local optimizer settings across methods. CPU work $C^{T,k}_m$ should be profiled for the model and hardware; if local epochs rather than steps are fixed, it also scales with dataset size.

For selected set $\mathcal{S}_k$,

$$w^{k+1} = w^k + \sum_{m \in \mathcal{S}_k} \alpha^k_m \Delta^k_m, \qquad \alpha^k_m = \frac{p_m}{\sum_{j \in \mathcal{S}_k} p_j}. \tag{7}$$

For UAV u, let $\mathcal{S}_{u,k}$ be its selected clients and $P_{u,k} = \sum_{m \in \mathcal{S}_{u,k}} p_m$. Its aggregated increment is

$$\Delta^k_u = \sum_{m \in \mathcal{S}_{u,k}} \frac{p_m}{P_{u,k}} \Delta^k_m. \tag{8}$$

The coordinator weights this by $P_{u,k} / \sum_v P_{v,k}$. Empty UAV groups are omitted. This exactly reproduces (7); equal averaging over UAVs generally does not.

Deterministic selection with selected-set normalization is not automatically an unbiased full-population update. Keep this aggregation rule fixed initially. With upload failures, aggregate successful clients only; if none succeed, retain $w^{k+1} = w^k$ and charge all attempted work.

2.2. Why client value is set-dependent

At a common model w, write $g_m = \nabla F_m(w)$ and $\bar{g} = \sum_m p_m g_m$. A diagnostic representativeness measure is

$$D(S) = \left\| \sum_{m \in S} \alpha_m(S) g_m - \bar{g} \right\|^2. \tag{9}$$

For a set utility estimate $\widehat{V}$, the marginal value is

$$\Delta V(m \mid S) = \widehat{V}(S \cup \{m\}) - \widehat{V}(S). \tag{10}$$

It can be negative and need not be submodular. Neither gradient novelty nor gradient similarity alone establishes usefulness.

Example 1: a useful minority direction. Nine equal-weight clients have gradient 0 and one has gradient -2. The population gradient is -0.2. Selecting the zero-gradient clients because they are closest to the population gradient gives no first-order update. This illustrates a limitation of interpreting similarity as usefulness; it is not an empirical claim about any particular published method.

Example 2: complementary clients. With equal weights and scalar gradients $(2, -2, 1)$, $\bar{g} = 1/3$. The pairs $\{1, 2\}$, $\{1, 3\}$, and $\{2, 3\}$ produce gradients 0, 1.5, and -0.5. Their squared matching errors are $1/9$, $49/36$, and $25/36$. The most representative pair contains very different individual gradients. Matching remains a proxy: multiple local SGD steps and final validation loss can behave differently.

2.3. History, stale information, and probing

Let $\mathcal{H}_k$ collect all permitted observations before round k. Store each client's last report, timestamp, model version, and sample count. Define

$$a_m^k = \tau_k - t_m^{\text{last}}. \tag{11}$$

Even stationary data yield stale gradients when the global model changes. Record both information age and model change, for example $\|w^k - w_m^{\text{last}}\|$.

A probe obtains minibatch loss and a gradient sketch:

$$y_m^k = \left(\widehat{F}_m(w^k), R\widehat{\nabla} F_m(w^k)\right) + \varepsilon_m^k, \qquad R \in \mathbb{R}^{d_s \times d}, \quad d_s \ll d. \tag{12}$$

The projection seed is shared so sketches are comparable. Noise represents sampling or measurement variation, not a formal differential-privacy guarantee. Computing the probe may require downloading the current model, which must be accounted for.

Maintain a belief or predictive distribution B_k^- over hidden client statistics:

$$B_k^+ = \text{Update}(B_k^-, y_{\mathcal{P}_k}^k). \tag{13}$$

An ensemble predictor is a practical option. Calibrate its uncertainty on held-out reports; arbitrary ensemble spread must not be described as a proven posterior variance. Unknown clients need a prior and an exploration opportunity, not a falsely precise zero contribution.

3. Communication, computation, and resource accounting

3.1. Decision variables and feasibility

Let $z_{m,u}^k$ indicate probing through UAV u and $s_{m,u}^k$ indicate training/uploading through it. Require

$$z_{m,u}^k, s_{m,u}^k \in \{0,1\}, \qquad \sum_u z_{m,u}^k \le 1, \quad \sum_u s_{m,u}^k \le 1. \tag{14}$$

Training without a fresh probe is allowed. Thus $s_m^k \le z_m^k$ is not imposed. Define $v_{m,u}^{P,k} = z_{m,u}^k$, $v_{m,u}^{T,k} = s_{m,u}^k$, and $v_m^{r,k} = \sum_u v_{m,u}^{r,k}$. For phases $r \in \{P, T\}$,

$$0 \le b_{m,u}^{r,k} \le B_u v_{m,u}^{r,k}, \qquad \sum_m b_{m,u}^{r,k} \le B_u, \tag{15}$$

$$f_m^{\min} v_m^{r,k} \le f_m^{r,k} \le f_m^{\max} v_m^{r,k}. \tag{16}$$

A measured mask $e_{m,u}^k$ enforces availability via $v_{m,u}^{r,k} \le e_{m,u}^k$. Rate and latency expressions apply only to active links with positive bandwidth; zero-bandwidth active schedules are infeasible.

3.2. Uplink and payload

With fixed transmit power and orthogonal spectrum pools,

$$R_{m,u}^{r,k} = b_{m,u}^{r,k} \log_2\left(1 + \frac{P_m h_{m,u}^k}{N_0 b_{m,u}^{r,k}}\right). \tag{17}$$

The base propagation option is $h_{m,u}^k = \beta_0 / \|q_u(\tau_k) - p_m(\tau_k)\|^2$ with positive altitude. More realistic models can replace it while retaining the formulation. State all units and convert dBm to W before evaluation. Rate (17) is an effective-rate approximation. An outage extension must specify coding, failures, and retransmissions consistently.

Let L_m^P be probe bits and L_m^T model-update bits, using fixed compression. For active links,

$$t_{m,u}^{r,k} = \frac{C_m^{r,k}}{f_m^{r,k}} + \frac{L_m^r}{R_{m,u}^{r,k}}. \tag{18}$$

Clients compute in parallel and upload on reserved orthogonal bands. The base scheduler does not reallocate bandwidth after a client finishes; this is conservative but reproducible.

3.3. Downlink, synchronization, and elapsed time

Let D_P^k and D_T^k be model-distribution times. Clients already holding w^k after probing do not download it again. Then

$$T_r^k = D_r^k + \max_{(m,u):\, v_{m,u}^{r,k}=1} t_{m,u}^{r,k}. \tag{19}$$

An empty phase has duration zero. Total duration is

$$T_k = t_{\mathrm{opt},P}^k + T_P^k + t_{\mathrm{opt},T}^k + T_T^k + t_{\mathrm{agg}}^k + t_{\mathrm{bh}}^k. \tag{20}$$

This assumes sequential phases. Overlap requires a critical-path calculation. Backhaul and aggregation must be modeled or calibrated, not implicitly free. For example, a sequential dedicated backhaul gives $t_{\mathrm{bh}}^k = \sum_{u:\, P_{u,k}>0} L_u^{\mathrm{agg}} / R_u^{\mathrm{bh}}$; simultaneous dedicated links use a maximum instead. Pick one protocol.

Optimizer latency is measured on the deployment hardware. If optimizer computations occur offline, do not add that offline time to every physical round; report it separately. Online searches consume elapsed time and possibly coordinator energy.

3.4. Energy, system boundary, and batteries

Under a dynamic CPU energy model,

$$E_m^{\mathrm{cmp},r,k} = \kappa_m C_m^{r,k} (f_m^{r,k})^2, \qquad E_{m,u}^{\mathrm{tx},r,k} = P_m \frac{L_m^r}{R_{m,u}^{r,k}}. \tag{21}$$

Calibrate κ_m; add static device power when material. Phase energy is the sum over active links. Total operational energy is

$$E_k = E_P^k + E_T^k + E_{\mathrm{DL}}^k + E_{\mathrm{agg}}^k + E_{\mathrm{bh}}^k + E_{\mathrm{flight}}^k + E_{\mathrm{opt}}^k. \tag{22}$$

For prescribed motion,

$$E_{\mathrm{flight}}^k = \sum_u \int_{\tau_k}^{\tau_{k+1}} P_u^{\mathrm{flight}}(t)\, dt. \tag{23}$$

Hovering yields $\sum_u P_u^{\mathrm{hover}} T_k$. Flight energy can depend on scheduling through elapsed time even if positions are not optimization variables. Report flight and radio/computation energy separately.

For physical node i, maintain

$$B_i^{k+1} = B_i^k - E_i^k \geq B_i^{\mathrm{reserve}}. \tag{24}$$

Coordinator energy enters the declared system total but not a UAV battery unless computation occurs onboard. Every cost is assigned to exactly one node ledger to prevent double counting.

Example 3: faster CPU is not always useful. Take $C = 10^8$ cycles, $\kappa = 10^{-28}$, $L = 2 \times 10^6$ bits, $R = 10^7$ bit/s, and $P = 0.1$ W. At $f = 10^9$ cycles/s, computation costs 0.1 s and 0.01 J; upload costs 0.2 s and 0.02 J. Total client time and energy are 0.3 s and 0.03 J. Doubling frequency gives 0.25 s and 0.06 J. If this client is not the slowest, the extra energy may not reduce round time.

4. Three-objective optimization problem

4.1. Nonanticipative two-stage control

Probing uses only information available beforehand:

$$a_P^k = (z^k, b^{P,k}, f^{P,k}) = \pi_P(\mathcal{H}_k). \tag{25}$$

After observing reports, training decisions are

$$a_T^k = (s^k, b^{T,k}, f^{T,k}) = \pi_T(\mathcal{H}_k, a_P^k, y_{\mathcal{P}_k}^k). \tag{26}$$

The second-stage action may depend on probe results, but not on unobserved current client statistics. Preselecting both stages without adapting after probing does not realize the full value of information.

4.2. Full-horizon objective

For a fixed K and initial checkpoint, consider

$$\min_{\pi_P,\pi_T} \mathbf{J}(\pi) = \left(\mathbb{E}[L_{\text{val}}(w^K)], \mathbb{E}\left[\sum_{k=0}^{K-1} E_k\right], \mathbb{E}\left[\sum_{k=0}^{K-1} T_k\right] \right). \tag{27}$$

Expectations cover declared channel, observation, and SGD randomness. Constraints include the model recursion, association, resources, batteries, and

$$1 \le \sum_m s_m^k \le N_{\max}, \qquad T_k \le T_{\max}. \tag{28}$$

The initial deterministic/block-rate model uses predicted realized latency. A stochastic extension can impose $\mathbb{P}(T_k \le T_{\max} \mid \mathcal{H}_k) \ge 1 - \epsilon$, but must define the uncertainty distribution and independently measure violation rates.

The base benchmark is designed to have at least one feasible participant per round. Operational states with no feasible participant require a declared recovery action, such as waiting for connectivity, with its time and energy charged. Do not silently discard those states.

Loss, energy, and time remain separate objectives. Use fixed normalization scales for search and report raw units. Changing normalization dynamically can distort preference comparisons.

4.3. Round-level approximation

Direct optimization of complete policies is difficult. At fixed snapshot S_k after any probing, Direction A solves

$$\min_{x \in \mathcal{X}(S_k)} \mathbf{j}_k(x) = \left(\mathbb{E}[L_{\text{val}}(w^{k+1}(x)) \mid S_k], E_k(x), T_k(x) \right), \qquad x = (s, b^T, f^T). \tag{29}$$

Completed probe costs are constant across training candidates but retained in episode accounting. A horizon-H extension predicts several rounds and executes only the first action.

A Pareto solution of (29) is not guaranteed to solve (27). One-round progress can neglect future coverage or battery depletion. Therefore evaluate both snapshot fronts and complete FL episodes.

4.4. Choosing an executed solution

One schedule must be selected from the feasible archive $\mathcal{A}_k$. A reproducible rule is

$$x_k^\star = \arg\min_{x \in \mathcal{A}_k} \widehat{L}_{\text{val}}(x) \quad \text{s.t.} \quad E_k(x) \le \bar{E}_k, \quad T_k(x) \le \bar{T}_k. \tag{30}$$

Budgets are fixed in advance. If no archived candidate meets them, use a predeclared minimum-violation fallback or a safe feasible schedule. Compare methods at common budgets or a common preference grid, not at operating points selected after looking at test accuracy.

5. Solution representation and reference evaluation

5.1. Mixed encoding and feasible decoding

Encode client association as $a_m \in \{0, \dots, U\}$, where 0 means inactive. Set $s_{m,u} = \mathbb{I}[a_m = u]$. Give active clients bandwidth keys ρ_m and CPU keys $\xi_m \in [0, 1]$:

$$b_{m,u} = B_u \frac{\exp(\rho_m)}{\sum_{j:a_j=u} \exp(\rho_j)}, \qquad f_m = f_m^{\min} + \xi_m (f_m^{\max} - f_m^{\min}). \tag{31}$$

Inactive clients receive zero resources. To permit unused bandwidth, add utilization $\gamma_u \in (0, 1]$ multiplying $b_{m,u}$. Under the base rate model full utilization is usually natural, but this should be checked if additional interference or radio costs are introduced.

Repair invalid associations and cardinality first, then resources and battery/deadline feasibility. Bound-preserving decoding does not automatically satisfy deadlines. When repair fails, retain constraint residuals and apply constraint dominance rather than silently declaring feasibility.

5.2. Analytical resource subproblem

For fixed bandwidth and a phase service-time bound d, an active client must satisfy

$$C_m/f_m + L_m/R_m \le d. \tag{32}$$

With only dynamic CPU energy, its lowest-energy feasible frequency is

$$f_m^\star = \max\left\{ f_m^{\min}, \frac{C_m}{d - L_m/R_m} \right\}, \tag{33}$$

provided $d > L_m/R_m$ and $f_m^\star \le f_m^{\max}$. Otherwise the assignment is infeasible. Sweeping d reduces the evolutionary dimension while exposing the energy–latency trade-off. This result concerns the stated fixed-deadline subproblem, not every extended energy model.

5.3. Reference evaluation and caching

Within one snapshot, fixed local SGD work is independent of bandwidth and CPU if all selected clients finish. A simulator can cache each client's reference update using identical checkpoint and random seeds, then aggregate different subsets. This is legitimate experimental acceleration. It is not a physical assumption that all devices train and transmit for free.

Charge cache construction to evaluation cost. Do not artificially inflate the computational difficulty by recomputing reusable local updates for every candidate. Across multiround branches, different aggregated models usually prevent reuse of later updates. This is where a genuine expensive-evaluation problem can emerge.

Highest fidelity still gives a finite-sample stochastic estimate. Repeats yield

$$\bar{L}(x) = \frac{1}{R}\sum_{r=1}^{R} L^{(r)}(x). \tag{34}$$

Use matched random seeds for paired comparisons when valid. Every candidate record must include snapshot and seed IDs. Evaluating candidates from different checkpoints and comparing their raw losses as one fitness landscape is invalid.

5.4. Separating controller information from evaluation

A reference evaluator may access simulator data to measure the true outcome of a candidate; the controller may not read those hidden data as free inputs. Offline optimization is explicitly labeled as such. For deployable tests, constrain the controller API to histories, measured links, and reports it has paid to acquire. Validation feedback, if obtained remotely, requires its own cost entry.

6. Algorithmic direction A: set-aware multiobjective evolutionary optimization

6.1. Initialization and operators

Direction A establishes a strong single-fidelity baseline. Initialize using random feasible schedules, channel-efficient schedules, representative-sketch schedules, and multiple cardinalities. Mix generic and guided proposals.

Discrete operators add, drop, swap, or reassign clients. Continuous operators mutate bandwidth and frequency keys. A complementarity-guided swap uses a proxy:

$$\widehat{D}(S \setminus \{i\} \cup \{j\}) < \widehat{D}(S), \tag{35}$$

where $\widehat{D}$ is computed from permitted sketches. The proxy guides proposal generation; reference learning quality determines survival in offline experiments. Random proposals preserve exploration when sketches are misleading.

For constraints $g_j(x) \le 0$, define

$$v(x) = \sum_j \frac{\max\{0, g_j(x)\}}{\sigma_j}, \tag{36}$$

with fixed dimensional scales $\sigma_j > 0$. Feasible candidates dominate infeasible ones; among infeasible candidates prefer smaller violation. Feasible survival uses Pareto rank and diversity preservation.

Algorithm 1 Direction A: reference-evaluated evolutionary scheduling

Require: Snapshot S_k, population size N, reference-evaluation budget
Generate and repair a mixed initial population $\mathcal{Q}$
Reference-evaluate candidates; initialize feasible archive $\mathcal{A}$
while budget remains **do**
    Select parents using feasibility, Pareto rank, and diversity
    Generate association and resource offspring
    Mix set-aware operators with random exploration
    Decode, repair, and compute analytical costs
    Reference-evaluate feasible offspring from the same snapshot
    Perform environmental selection and update $\mathcal{Q}$
    $\mathcal{A} \leftarrow \mathrm{ND}(\mathcal{A} \cup$ verified feasible offspring$)$
end while
Select one schedule using the declared deployment preference

Table 2
Evaluation fidelity versus physical probing. These are different mechanisms.

Mechanism	Information produced	Cost accounting
Sketch predictor	Estimated outcome of a candidate	Optimizer inference time/energy
Reduced simulator rollout	Approximate learning outcome	Search evaluation budget
Reference simulator rollout	Full protocol outcome estimate	Search and audit budgets
Physical client probe	Newly observed client statistic	Download, compute, upload, waiting
Executed FL round	Real model update and feedback	Full operational ledger

6.2. Hypotheses

Test whether set-aware operators improve the reference-verified front at equal evaluation cost and whether analytical CPU decoding saves search effort. Compare with generic NSGA-II, a decomposition-based MOEA, random feasible search, and greedy/data-diversity scheduling with the same allocator. Merely applying a known MOEA to UAV-FL is not sufficient evidence of methodological novelty.

7. Algorithmic direction B: multi-fidelity evolutionary optimization

7.1. Main idea and evaluation hierarchy

Direction B allocates search effort among inexpensive predictions and expensive training-based evaluations. Candidate features can include selected-client sketch moments, estimated matching error, selected population mass, uncertainty, cardinality, information ages, channels, and resources. Avoid arbitrary client indices for cross-topology generalization.

Define learning evaluations

$$\widehat{L}_1(x) = \text{calibrated sketch/history prediction}, \tag{37}$$

$$\widehat{L}_2(x) = \text{reduced-work or reduced-validation rollout}, \tag{38}$$

$$\widehat{L}_3(x) = \text{full-work reference validation}. \tag{39}$$

Physical energy and time are often analytically computable at all levels. There is no benefit in approximating them solely to create more fidelity levels. When Level 2 runs fewer SGD steps, its analytical physical costs must still describe the target full-work schedule, not the shortened evaluation procedure.

Lower fidelity can reverse candidate ranking. Before developing promotion rules, measure runtime, bias, rank correlation, and calibration against the reference. If cached reference evaluation is already cheap, multi-fidelity may not help.

Algorithm 2 Direction B: cost-aware multi-fidelity evolutionary search

Require: Snapshot, total evaluation budget B, audit reserve B_{audit}
1: Generate a diverse initial set and reference-evaluate a seed subset
2: Fit cross-fidelity calibration and initialize reference archive
3: **while** search cost $< B - B_{\text{audit}}$ **do**
4: Generate and repair candidates with Direction A operators
5: Compute analytical objectives and Level-1 learning predictions
6: Update the provisional archive
7: Select promotions using acquisition plus reserved exploration
8: Evaluate selected candidates at chosen fidelities
9: Update calibration data and reference-verified archive
10: **end while**
11: Audit final candidates using reserved independent reference evaluations
12: Return the consistently evaluated feasible nondominated set

7.2. Cross-fidelity calibration

One possible discrepancy model is

$$L_3(x) = a_\ell \widehat{L}_\ell(x) + b_\ell + d_\ell(\phi(x)) + \epsilon_\ell, \qquad \ell \in \{1, 2\}. \tag{40}$$

Train on paired evaluations from the same snapshot. An ensemble gives a mean $\mu_3(x)$ and calibrated uncertainty $s_3(x)$. Across rounds, condition on snapshot descriptors or predict improvement relative to the current validation loss. Mixing raw losses from different training stages without conditioning corrupts the predictor.

A conservative estimate $\mu_3(x) + \beta s_3(x)$ can be used for final minimization decisions. However, conservatism alone can exclude unknown clients forever; evaluation and probing policies need exploration.

7.3. Cost-aware promotion

For measured evaluation costs $c_\ell(x)$, enforce

$$\sum_{(x,\ell)\in Q_{\text{eval}}} c_\ell(x) \le B_{\text{search}}. \tag{41}$$

Use a declared unit such as GPU-seconds or wall time; record parallelism. One proposed promotion heuristic is

$$A(x, \ell) = \frac{\widehat{\mathbb{P}}(x \text{ changes the reference front})\, \widehat{\Delta U}(x, \ell)}{c_\ell(x) + \epsilon}. \tag{42}$$

Here $\widehat{\Delta U}$ predicts reduction of uncertainty relevant to front membership. Estimate the probability by sampling the calibrated loss predictor with analytical energy/time. This is a research heuristic, not a proven optimal acquisition policy.

A simpler implementable alternative is optimistic hypervolume improvement per cost. For normalized minimization objectives, compute an optimistic point using $\mu_3 - \beta s_3$ for loss and exact analytical energy/time, evaluate its potential hypervolume gain, and divide by promotion cost. Retain an explicit exploration fraction; otherwise inaccurate confidence estimates can eliminate useful candidates permanently. Compare the heuristic with random promotion, uncertainty-only promotion, and a fixed fidelity schedule.

7.4. Verified archives and final audits

Maintain $\mathcal{A}_{\text{pred}}$ for provisional surrogate-guided candidates and $\mathcal{A}_{\text{ref}}$ for common-protocol reference evaluations. Only the latter supports final Pareto-quality claims. Do not mix predicted and reference loss values in headline hypervolume calculations.

Reserve audit resources before search. Re-evaluate reported solutions with an independent common reference protocol, then recompute nondominance. Report search and audit cost separately, including discarded evaluations.

Algorithm 3 Direction C: probing with adaptive second-stage scheduling

Require: Initial model, batteries, histories, predictor, common deployment budgets

for $k = 0, \dots, K-1$ **do**
  Observe permitted physical state and update stale-information features
  Generate candidate probe sets including the empty set
  Estimate net information value under current resource constraints
  Execute a feasible probe plan and charge all physical costs
  Update client reports, timestamps, and belief uncertainty
  Reoptimize training selection and resources with the new information
  Execute one schedule chosen by the predeclared Pareto preference
  Aggregate updates and advance the physical clock
  Update batteries and accessible validation/learning feedback
end for

7.5. Offline versus online interpretation

Offline branching can evaluate many counterfactual schedules. A physical online controller generally cannot perform full local training for many alternative sets without paying for it. The deployable version uses simulator-trained models, permitted history, and charged probes; it executes one schedule and uses accessible feedback to update its predictor.

Online trial training, if used, is an actual operation and must appear in the operational ledger. If labels or validation feedback are delayed, state the feedback delay and do not pretend instantaneous full validation is available. Monitor calibration under deployment shift and define a fallback to a conservative feasible schedule when predictions become unreliable.

8. Algorithmic direction C: adaptive probing and online recourse

8.1. Value of acquiring information

Uncertainty is useful only if resolving it could change the executed schedule. A highly uncertain client with no feasible uplink may not justify probing.

For a fixed deployment preference λ, define a normalized scalar cost

$$C_\lambda(a,\theta) = \lambda_L \widetilde{L}(a,\theta) + \lambda_E \widetilde{E}(a) + \lambda_T \widetilde{T}(a), \tag{43}$$

where θ denotes hidden learning statistics. Holding the second-stage action domain fixed for the conceptual comparison, expected sample information value is

$$\mathrm{EVSI}(P) = \min_a \mathbb{E}[C_\lambda(a,\theta) \mid \mathcal{H}_k] \tag{44}$$

$$- \mathbb{E}_{y_P}\left[\min_a \mathbb{E}[C_\lambda(a,\theta) \mid \mathcal{H}_k, y_P]\right]. \tag{45}$$

Subtract the normalized probe cost to estimate net value. In the actual constrained problem, probe execution also reduces remaining budgets and changes elapsed time; account for these effects during recourse.

Approximate (45) with a small set of belief scenarios and candidate training schedules. Include the empty probe set. This approximation is a proposed component requiring ablation, not an established guarantee.

8.2. Online procedure

Rerunning the training scheduler after probing is essential. Keeping the original training set fixed tests reporting overhead, not meaningful adaptive recourse.

8.3. Optional nonstationary extension

After stationary experiments, introduce new classes, gradual distribution shifts, or intermittent connectivity. At fixed mission horizon T_{mission}, replace terminal loss by

$$J_L^{\mathrm{online}} = \frac{1}{T_{\mathrm{mission}}} \mathbb{E}\left[\int_0^{T_{\mathrm{mission}}} L_{\mathrm{val},t}(w(t))\, dt\right]. \tag{46}$$

Table 3
Staged benchmark matrix.

Factor	Initial study	Stress test
Scale	$M = 6, 20$; $U = 2$	$M = 50, 100$; $U = 3, 5$
Data/model	Fashion-MNIST, small CNN	CIFAR-10, larger CNN
Heterogeneity	IID; Dirichlet $\alpha = 1, 0.1$	Geographically correlated classes
Rare data	Controlled minority classes	Minority clients on weak links
Data quality	Clean datasets	Declared fraction of label noise
Observability	Fixed fresh/historical reports	Costly probes and stale reports
Dynamics	Stationary data, prescribed motion	Physical-time distribution shifts
Evaluation	One-round cached reference	Multiround branching rollout

This measures sustained quality instead of final recovery alone. Trigger shifts using common physical times, not FL round numbers, because algorithms can have different round durations. Continual learning and personalized modeling are separate design choices and should not be added before the scheduling contribution is understood.

9. Benchmark construction and experimental protocol

9.1. Research questions

RQ1: Does set-aware scheduling improve the loss–energy–time trade-off over channel-only and generic evolutionary methods?

RQ2: Does multi-fidelity evaluation improve independently verified solution quality at equal actual search cost?

RQ3: Does adaptive probing improve complete FL episodes after its physical overhead is counted?

RQ4: Which regimes cause failures through rare-client exclusion, noisy data, stale reports, or fidelity rank reversal?

9.2. Scenario construction

The following are proposed starting configurations, not measured system specifications. Fix the total number of training examples when varying client count, or label a growing-data study explicitly. Use independent data partitions and mobility traces. Create paired scenarios moving the same rare-data client from a good to a poor channel, to isolate statistical–physical coupling. State the class imbalance used in both target validation and test sets.

9.3. Baselines and fairness

Use three baseline groups:

1. **Scheduling:** random participation, channel greedy, an Oort-style utility policy, a gradient-diversity policy, and a gradient-similarity policy. Identify adaptations rather than claiming exact reproductions.
2. **Optimization:** random feasible search, single-fidelity NSGA-II, a decomposition-based MOEA, and the same proposed operators without multi-fidelity.
3. **Information:** no new probes, periodic probes, random probes, uncertainty-only probes, information-value probes, and a fully observed oracle.

Deployable methods receive the same permitted information; extra statistics carry acquisition cost. The oracle is a diagnostic comparator, not a fair deployable competitor. Use the same resource allocator when isolating client selection and the same selection when isolating allocation.

Equalize actual optimizer time or profiled evaluation cost, not only fitness-call counts. Include cache construction, model fitting, promotion, and reference audits. Match worker counts and hardware. Tune on separate scenarios; hold out data partitions, mobility traces, and network sizes where practical.

Table 4
Suggested implementation responsibilities.

Module	Responsibility
`fl_engine`	Immutable checkpoints, local SGD, aggregation, validation
`network_model`	Eligibility, rates, downlink/backhaul, phase times
`energy_model`	Node-level cost ledger, hovering, batteries
`schedule`	Encoding, decoding, repairs, constraint residuals
`evaluators`	Fidelity wrappers, seeds, caches, measured search costs
`moea`	Operators, selection, provisional and verified archives
`belief_probe`	Reports, uncertainty, probe choice and recourse
`experiments`	Scenario manifests, paired runs, audits and metrics

9.4. Metrics and statistical reporting

For snapshot fronts, report reference-verified hypervolume using common normalization and a fixed dominated reference point. Use IGD only with a credible common reference set; label a pooled empirical reference as approximate and test sensitivity.

For complete FL runs, report final test loss/accuracy, macro-F1, rare-class recall, energy/time to a common accuracy target, battery depletion, deadline violations, probe bytes and energy, and optimizer latency. Explicitly report runs that do not reach the target. Under drift, measure time-integrated loss and recovery delay.

Use at least five independent seeds for pilot work and aim for ten or more paired runs for main comparisons, guided by observed variance and compute cost. Report mean, dispersion, and paired confidence intervals. Tuning trials are not independent test runs. Inspect full trajectories; final accuracy alone can conceal long periods of poor performance.

9.5. Ablations and failure diagnostics

- Set-dependent versus independent client scoring.
- Generic versus guided mutation; analytical versus evolved CPU allocation.
- Single fidelity versus fixed promotion versus adaptive promotion.
- Cross-fidelity calibration and exploration reserve on/off.
- No, periodic, uncertainty-only, and information-value probing.
- Full operational accounting versus deliberately omitted costs as a diagnostic only.
- One-round versus short-horizon search under equal compute budgets.

Measure fidelity rank reversals, prediction calibration across training stages, promotion cost per verified gain, and rare/noisy client selection frequencies. Explain negative results, especially when one-round reference caching eliminates the intended evaluation bottleneck.

10. Implementation roadmap and verification exercises

10.1. Software modules

An evaluation record contains snapshot ID, candidate hash, fidelity, partition ID, random seed, objectives, constraint residuals, evaluation cost, and verification status. Cache keys include model version and local-training configuration. Keep hidden simulator truth separate from the controller interface.

10.2. Student work packages

WP1. Six-client physical model. Implement links, allocations, duration, energy, and batteries. Deliver a hand-checked instance and restricted subset enumeration.

WP2. FL integration. Add local SGD and weighted UAV aggregation. Deliver reproducible snapshots and numerical equivalence of hierarchical and flat aggregation.

WP3. Direction A. Implement generic and set-aware search. Deliver consistently evaluated fronts and end-to-end runs under one common deployment rule.

WP4. Fidelity pilot. Measure cached reference costs, reduced-rollout costs, rank correlations, and calibration. Deliver evidence that multi-fidelity is useful, or revise the setting.

WP5. Direction B. Add promotion, calibration, exploration, and independent auditing. Deliver equal-cost comparisons and ablations.

WP6. Direction C. Enforce partial observation and add charged probing with recourse. Deliver stale-information and rare-data experiments.

Different students can own modules, but use a common configuration schema and cost ledger. Integrate after each work package. Do not develop multiple incompatible simulators and combine them only at the end.

10.3. Verification exercises

1. Reproduce the arithmetic in Examples 1–3.
2. Verify inactive clients receive zero resources and incur no local-training cost.
3. Check association, bandwidth, frequency, and battery constraints after every decode.
4. Compare hierarchical and flat aggregation to numerical precision.
5. Confirm candidate evaluation never changes the source checkpoint.
6. Verify identical seeds reproduce local updates and evaluation values.
7. Ensure probing and training do not double-charge the same model download.
8. Check that a deployable controller cannot access unqueried current gradients.
9. Confirm longer hovering duration increases flight energy.
10. Re-evaluate every reported Pareto point at the same reference fidelity.
11. For six clients, enumerate all nonempty subsets with fixed allocation and compare with the search archive. Label this as an exact restricted benchmark, not an exact solution of the full continuous problem.
12. Use synthetic known objectives to test dominance and archive updates independently of FL.

10.4. Decision gates and contribution boundaries

Proceed to Direction B only if profiling shows an evaluation bottleneck and a useful fidelity hierarchy. Proceed to Direction C only after the scheduler works with a fixed information set. If probing overhead exceeds its gains, characterize when historical reports are sufficient.

Potential contributions include consistent information/resource modeling, set-aware operators, cost-aware fidelity promotion with verified archives, and evidence about stale-information failure modes. These are proposed contributions, not established novelty or convergence claims. The eventual paper must compare against the closest UAV-FL, client-selection, and multi-fidelity optimization literature.

Recommended first experiment. Use two UAVs, six clients, fixed local work, and one initial checkpoint. Place the rare-class client on a weak link, enumerate a restricted space, and measure true validation trade-offs. Compare generic with set-aware search at equal reference cost. Introduce surrogate evaluation and adaptive probing only after this controlled experiment is reliable.